CASPEr@night — Agilent/VnmrJ Quickstart

(SIU Carbondale — 400 MHz DD2, VnmrJ 3.2)

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1 What this is

This guide covers one specific session: a receive-only spin-noise measurement on the SIU Carbondale 400 MHz Agilent DD2 (VnmrJ 3.2, 5 mm HCN probe, room temperature). It is the Agilent leg of the CASPEr@night network — the same physics protocol the Bruker fleet runs, recording the Guéron absorption dip of water on a room-temperature probe — acquired by hand with ordinary VnmrJ tools (`s2pul`, `go`, `svf`) and converted into a network bundle afterward on any laptop. Nothing in the session pulses the sample above tune-and-match power: the calibration steps use $\sim$1° tips at your ordinary observe power, and the noise blocks use `pw=0` at the bottom of the transmitter power scale. One thing to know going in: **yours is the first real Agilent data this software will ever see.** The converter has passed synthetic-session and file-format tests against the documented VnmrJ file layout, but it has never parsed a byte from a real spectrometer — so if your files make it choke, that is exactly the data we need. Ugly output is useful output, and Section 4 says what to send. You never need the GitHub repository, git, or a README: this PDF plus the `config.json` attached to the same e-mail is everything.

What you need

1. **Any laptop or desktop with Python 3.6 or newer** — macOS, Linux, and Windows all work; nothing is installed on the spectrometer. Check with:
   ```
   python3 --version
   ```
2. **The software — one download, no git.** Open
   ```
   https://github.com/CASPEr-Night/spin-noise-network/releases/latest
   ```
   and under *Assets* click **Source code (zip)**. Unzip it anywhere.
3. **The `config.json` from John's e-mail** — drop it into the unzipped folder at exactly `uploader/config.json` (next to `upload_bundle.py`). It is your facility's upload key — treat it like a password.
4. **A 5 mm tube of 90% H_2O / 10% D_2O, lightly doped** — a retired shim/lineshape sample is fine, as long as you know (or can best-guess) the composition. The H_2O fraction *is* the measurement.
5. **About 1 hour of instrument time** — roughly 15 min of setup you attend, then $\geq$30 min of noise blocks the console mostly runs itself.

2 At the spectrometer

Everything below is standard VnmrJ operation — no macro, nothing installed on the console. Save every experiment with `svf` into one session directory, one `.fid` per step; the leading number in each save name is what the converter reads (the rest of the name is free).

2.1 Setup (nothing new here, ∼15 min)

Insert the water tube, set or record the temperature (298 K if VT is off and the bore is at room temperature), and let it equilibrate a few minutes. Tune and match ^{1}H by your normal manual wobble procedure — no autotune on this probe, and none needed. Shim as usual. Lock is your choice, on or off — you will record which in the questionnaire. Calibrate the ^{1}H 90° pulse or take the probe-file value, and **write down pw90 and tpwr** — the questionnaire wants both. Start from your standard ^{1}H water setup in `s2pul` and leave `sw` at your usual value throughout.

2.2 Gain ladder (∼5 min)

Four one-scan 1Ds at ascending receiver gain, with a tiny flip so nothing saturates. Set `pw` to pw90/90 — 0.11 μs for a 10 μs pw90, about a 1° tip — never use `gain='n'` (autogain), and wait for each acquisition to finish before saving:

```
pw=0.11
nt=1
gain=0
go
svf('10_sn_ladder_a')
gain=20
go
svf('14_sn_ladder_b')
gain=40
go
svf('15_sn_ladder_c')
gain=60
go
svf('16_sn_ladder_d')
```

Documentation says `gain` runs 0–60 dB on this hardware family, but nobody has verified the DD2's legal values or overflow point for us: **if the console rejects `gain=60` or the receiver overflows, step down in 5 dB until it runs clean**, note the values you actually used in the questionnaire, and read every `gain=60` below as that ladder-max value.

2.3 Opening reference (∼1 min)

One more tiny-flip 1D at the ladder-max gain, with a 2 s record:

```
at=2
gain=60
go
svf('11_sn_ref_open')
```

2.4 Noise blocks (≥30 min — the point of the session)

No pulse at all — these lines are the receiver-only idiom of Agilent's own `cryo_noisetest` macro, with `tpwr=-16` the bottom of the standard transmitter power scale:

```
pw=0
tpwr=-16
nt=1
ss=0
pad=0
wshim='n'
alock='n'
gain=60
at=60
go
svf('12_sn_noise')
```

Then repeat until you have at least 30 minutes of noise data in total, counting up from 17 (13 is reserved for the closing reference):

```
go
svf('17_sn_noise')
go
svf('18_sn_noise')
```

Three places your judgment is needed — each is an unverified point we ask you to settle:

- **Record length.** Try `at=60` (30 blocks reach 30 min); if the console refuses — a maximum-points ceiling we have not measured on the DD2 — halve `at` until it accepts and note the largest value that worked. If it caps near 10 s, do not grind out 180 saves: take about 20 blocks, note the cap, and stop — on this first session a parseable short set beats a long one.
- **Transmitter silence.** Whether the DD2's transmitter chain is measurably silent at `pw=0 tpwr=-16` has never been checked. If the noise floor shows something odd — a carrier spike, a ridge — do not chase it: finish the session and describe it in the questionnaire notes.
- **Save names.** If `svf` objects to a name starting with a digit, save under any name and rename the `.fid` directories to the `NN_...` pattern when you copy them off (only the leading number is parsed) — and note it for us.

2.5 Closing reference (∼1 min)

Restore the reference parameters — `tpwr=57` below standing in for your calibrated value from setup — and save as 13:

```
pw=0.11
tpwr=57
at=2
go
svf('13_sn_ref_close')
```

That is the whole session. A VnmrJ macro that runs all of it unattended (`spin_noise_run.mac`) exists in draft — once this first manual session validates the converter, we will ask for one more hour of bench time to try it.

3 Back at your desk

3.1 Copy the data

Copy the whole session directory — every `NN_*.fid` folder — onto the laptop, any path. A USB stick is fine; the console never needs internet access.

3.2 Fill in `answers.json`

In the unzipped `spin-noise-network` folder, create `answers.json` with the content below — it is pre-filled for SIU, so only the `FILL IN` lines (and `"locked"`, `"p90_us"`, and `"vt_setpoint_k"` if yours differed) are yours to edit:

```
{
  "vendor": "agilent",
  "run_mode": "external-acquisition",
  "facility": {
    "institution": "Southern Illinois University",
    "city": "Carbondale, Illinois",
    "country": "USA",
    "facility_slug": "siu-carbondale",
    "contact_email": "bgoodson@chem.siu.edu",
    "contact_consent": true
  },
  "sample": {
    "description": "FILL IN (e.g. retired lineshape sample, 90% H2O / 10% D2O, Gd-doped)",
    "h2o_fraction_pct": 90,
    "d2o_pct": 10,
    "additives": "FILL IN (dopant, if known)",
    "tube_od_mm": 5,
    "sample_volume_ul": 550,
    "vt_setpoint_k": 298
  },
  "environment": {
    "locked": false,
    "operator_notes": "FILL IN (date; anything unusual during the run)"
  },
  "spectrometer": {
    "probe_type": "RT",
    "console": "Agilent DD2 400",
    "probe_string": "5 mm HCN"
  },
  "instrument": {
    "vnmrj_version": "3.2",
    "spectrometer_model": "Agilent DD2 400",
    "field_state_notes": "FILL IN (lock on/off during the noise blocks; z0 stable?)"
  },
  "calibration": {
    "p90_us": 10.0,
    "p90_power_db_or_w": "FILL IN (tpwr, dB)",
    "topshim_ok": false,
    "rg_ladder": [
      { "expno": 10, "rg": 1.0,    "tip_deg": 1.0 },
      { "expno": 14, "rg": 10.0,   "tip_deg": 1.0 },
      { "expno": 15, "rg": 100.0,  "tip_deg": 1.0 },
      { "expno": 16, "rg": 1000.0, "tip_deg": 1.0 }
    ]
  },
```

```
  "experiments": [
    { "expno": 10, "role": "rg_ladder" },
    { "expno": 14, "role": "rg_ladder" },
    { "expno": 15, "role": "rg_ladder" },
    { "expno": 16, "role": "rg_ladder" },
    { "expno": 11, "role": "reference_open" },
    { "expno": 12, "role": "noise" },
    { "expno": 17, "role": "noise" },
    { "expno": 18, "role": "noise" },
    { "expno": 13, "role": "reference_close" }
  ]
}
```

Two mechanical notes: add one `{ "expno": NN, "role": "noise" }` line per noise block you saved beyond 18, and the ladder `"rg"` values are $10^{\mathrm{dB}/20}$ of the four gains — pre-filled for 0/20/40/60 dB, so correct them if your ladder used different values.

3.3 Pack, check, upload

Three commands, run from inside the unzipped `spin-noise-network` folder:

```
python3 packer/pack_bundle.py --vendor agilent --answers answers.json
      /path/to/the/data/folder
```

Success looks like: the packer validates the session and prints the path of one `spinnoise_siu-carbondale_*.zip` — and if anything is missing, it lists exactly what, in plain text, so you fix it and re-run.

```
python3 uploader/upload_bundle.py --selftest spinnoise_*.zip
```

Success looks like: the last line reads `RESULT: PASS` — entirely local, no network involved yet.

```
python3 uploader/upload_bundle.py spinnoise_*.zip
```

Success looks like: the last line reads `RECEIPT: <id>` — the server has your bundle, checksum-verified on arrival; uploads are chunked and resumable, so if the connection drops, re-run the same command and it picks up where it left off.

4 If anything goes wrong

One rule: send it, don't fix it

Send the **exact error text** (copy-paste, not a paraphrase) and the file it choked on — the offending `NN_*.fid` folder zipped, or the bundle zip — to jwbquantum@gmail.com. Troubleshooting data is the point of this first run: the converter has never met a real Agilent file, and your session is how it becomes trustworthy. Nothing you can do here damages anything — the software only reads your data, the uploader never deletes your zip, and the worst case at every step is re-running one command. If the whole chain fights you, zip the entire session directory and e-mail that (a share link is fine if it is large); we take it from there.

Contact: John W. Blanchard (jwbquantum@gmail.com) — CASPEr@night software v0.4.0 — August 28, 2026.
Your raw data remain yours, and contributing facilities are co-authors on network publications their data enter.